

```
1 import java.util.*;
2 class Saddle_Point
3 {
4     static int a[][] , r, c; //declaing the data members
5
6     Saddle_Point(int row, int column)
7     {
8         r=row;
9         c=column;
10        a=new int[r][c];
11    }
12
13    void input()
14    {
15        Scanner sc=new Scanner(System.in);
16        System.out.println("PLEASE ENTER THE ELEMENTS ");
17        for (int i = 0; i < r; i++)
18        {
19            for (int j = 0; j < c ; j++)
20            {
21                a[i][j]=sc.nextInt();// taking input from the user
22            }
23        }
24    }
25
26    void display()
27    {
28        System.out.println();
29        for (int i = 0; i < r; i++)
30        {
31            for (int j = 0; j < c ; j++)
32            {
33                System.out.print("\t"+a[i][j]); //dispkaying each element of the
matrix
34            }
35            System.out.println();
36        }
37    }
38
39    boolean SaddlePoint()
40    {
41        // Process all rows one by one
42        for (int i = 0; i < r; i++)
43        {
44            // Find the minimum element of row i.
45            // Also find column index of the minimum element
46            int min_row = a[i][0], col_ind = 0;
47            for (int j = 1; j < c; j++)
48            {
49                if (min_row > a[i][j])
50                {
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51         min_row = a[i][j];
52         col_ind = j;
53     }
54 }
55
56 // Check if the minimum element of row is also
57 // the maximum element of column or not
58 int k;
59 for (k = 0; k < c; k++)
60
61     // Note that col_ind is fixed
62     if (min_row < a[k][col_ind])
63         break;
64
65 // If saddle point is present in this row then
66 // print it
67 if (k == r)
68 {
69     System.out.println("Value of Saddle Point " + min_row);
70     return true;
71 }
72 }
73 return false;
74 }
75
76 public static void main()
77 {
78     Scanner sc=new Scanner(System.in);
79     System.out.println("ENTER THE NUMBER OF ROWS (BETWEEN 3 & 6 INCLUSIVE) ");
80     r=sc.nextInt();
81     System.out.println("ENTER THE NUMBER OF COLUMNS (BETWEEN 3 & 6 INCLUSIVE) ");
82     c=sc.nextInt();
83     Saddle_Point ob=new Saddle_Point(r,c); //creating an object of class
84     ob.input();
85     ob.display();
86     boolean k=ob.SaddlePoint();
87     if(k==false)
88         System.out.println("THE SADDLE POINT DOES NOT EXIST");
89 }
90 }
91
92
```